

Inside Classifier Tokenomics: A Practical Guide to Choosing Between LLMs, TypeSafe.ai JEV, SLMs, and Open-Source Encoders

How state length, question count, model quality, GPU utilization, and latency change the right classifier architecture

13	5	2
embedded decision charts	variables that move the answer	operating perspectives

For data scientists choosing a practical model and enterprise teams deciding when to operate a shared platform.

Mark Austin | September 2026

Guide structure

1. One classification task, five execution patterns
2. What JEV is - and what public evidence establishes
3. The quality ladder from this benchmark
4. Tokenomics: the variables that actually move cost
5. Latency: resource time is not response time
6. The data scientist's decision path
7. The enterprise platform team's decision path
8. When to move off a cloud API
9. What we are testing now
10. A pragmatic reference architecture
11. Bottom line and visual editorial notes

Visual logic

Architecture comes before the leaderboard. Quality evidence comes before cost. State length and question count come before utilization. Deployment and operating-model choices come last, after the reader understands what each system actually computes.

Opening

Too many options, too many curves, and too many claims. Should a classifier be a frontier LLM, a small language model, a hosted "System One" API, a zero-shot encoder, calibrated thresholds, or a fully trained model? Should a data scientist call an API, download a checkpoint, or ask the platform team to buy H100s? And what is JEV doing differently enough to change the answer?

The uncomfortable truth is that "which model is best?" is usually the wrong question. The useful answer changes with four variables: state length, number of questions, quality target, and paid hardware utilization. Latency adds a fifth constraint because the cheapest saturated system may be the worst interactive system once requests queue.

This article maps the choices from two perspectives: the practitioner shipping one use case and the enterprise team operating a shared platform for many teams.

The examples use transcript classification because it gives us a concrete workload, but the framework applies to any problem with a large shared state and multiple decisions: document processing, tool routing, safety gates, claims review, compliance checks, or product recommendations.

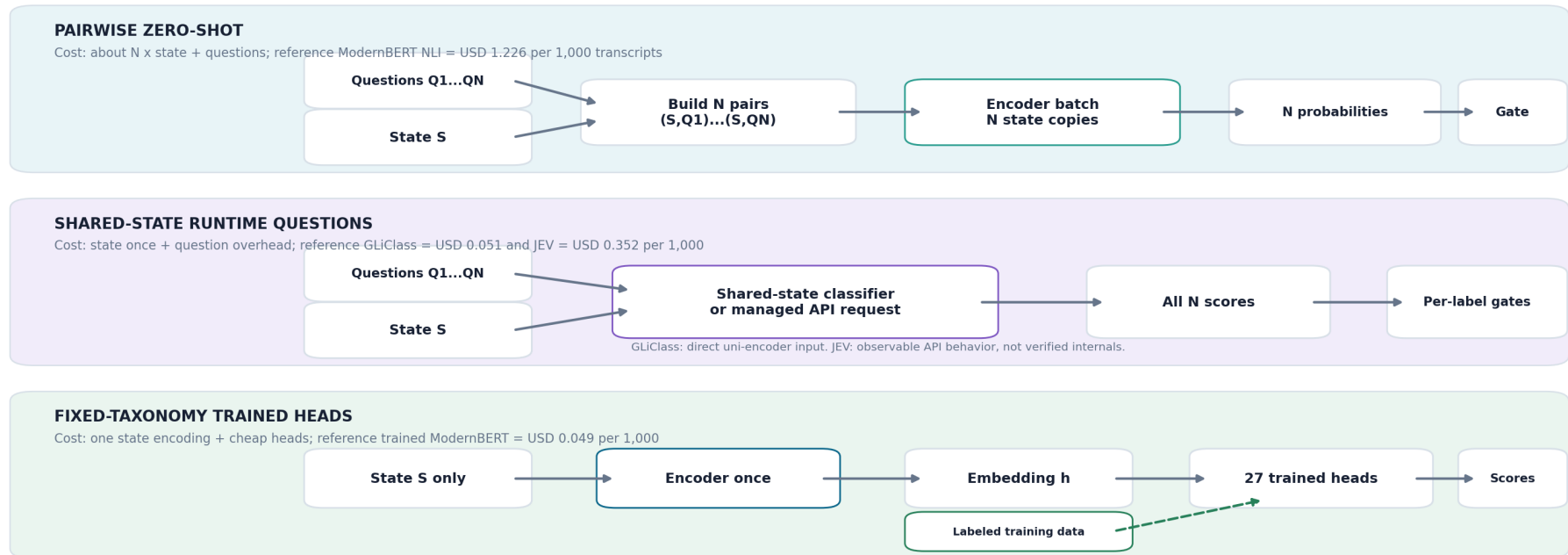
1. One classification task, five execution patterns

- Pairwise zero-shot encoder: repeat the state for every arbitrary question.
- Shared-state runtime-label encoder: serialize one state with many labels.
- Fixed-taxonomy trained heads: encode the state once; label semantics live in weights.
- Hosted shared-state API: submit one state plus arbitrary typed questions.
- Single-call LLM: prompt for all decisions and structured output.

Include Mermaid diagrams showing the state path, question path, and cost driver for each.

FIGURE 1 | THREE WAYS TO COMBINE A SHARED STATE WITH CLASSIFICATION QUESTIONS**Three ways to combine state and classification questions**

Reference: 6k state tokens x 25 questions, 1,000 transcripts, 100% paid H100 utilization; JEV remains usage-priced



Calibration changes gates after scoring; it does not change whether the state is repeated, shared, or encoded once for fixed learned heads.

Figure 1. The central architectural distinction: pairwise zero-shot repeats the state, shared-state systems accept runtime questions without repeating the state at the product boundary, and trained heads encode the state once but fix the taxonomy in advance.

2. What JEV is - and what the public evidence establishes

- Noul, Choice, and Score primitives.
- One shared state plus arbitrary runtime questions.
- Current documented context, pricing, and parallel evaluation behavior.
- Clear distinction between observed/documented interface behavior and undisclosed architecture.
- Why a GLiClass compatibility layer is useful without claiming it reproduces JEV.

3. The quality ladder from this benchmark

- Raw zero-shot encoders.
- Validation-calibrated zero-shot encoders.
- Trained ModernBERT heads.
- JEV Noul and Choice.
- Luna and Sol.
- Highlight: calibration closed more than half the ModernBERT zero-shot-to-trained gap while preserving runtime question flexibility for the known labels.

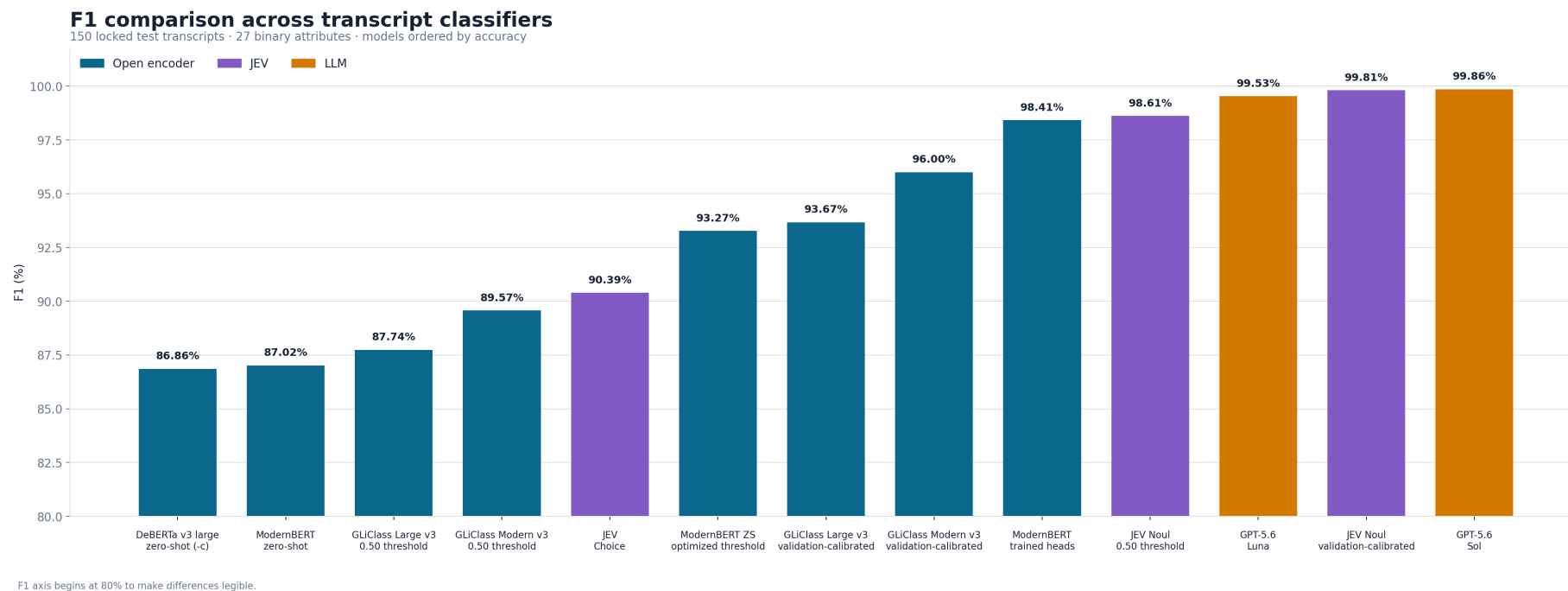
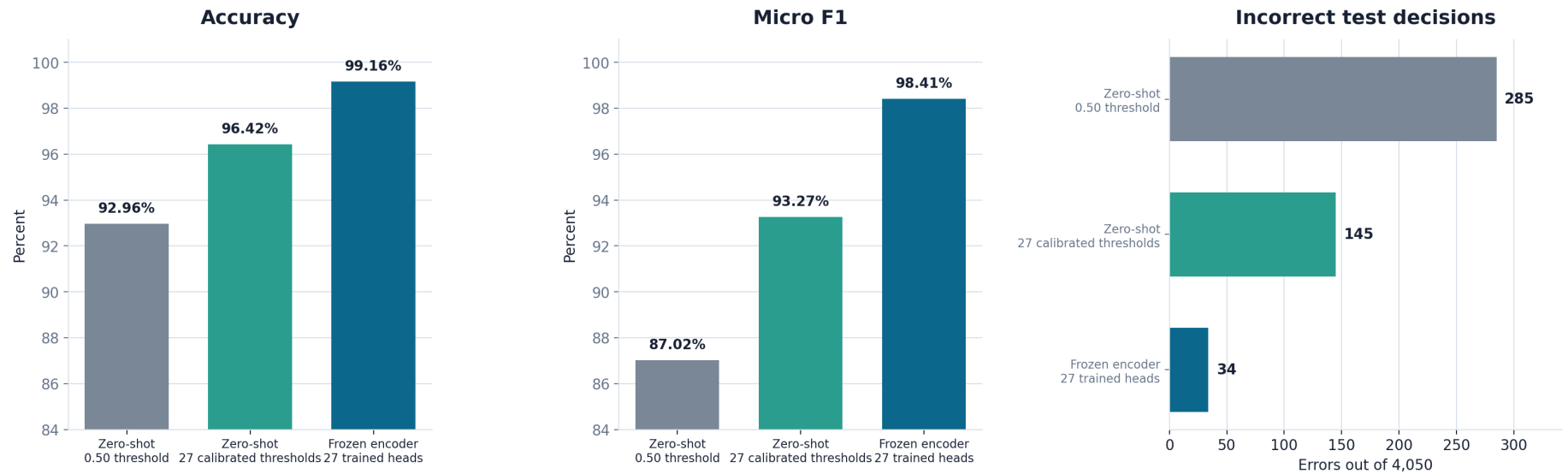
FIGURE 2 | F1 COMPARISON ACROSS THE TESTED TRANSCRIPT CLASSIFIERS

Figure 2. Held-out micro F1 across 4,050 binary decisions. F1 is the primary comparison because the positive and negative classes are imbalanced.

FIGURE 3 | THRESHOLD CALIBRATION VERSUS TRAINED MODERNBERT HEADS**Threshold calibration captures much of training's gain without changing model weights**

ModernBERT-large · same 4,050 locked test decisions · calibration closes 55.8% of the accuracy gap and 54.8% of the F1 gap



Calibration changes only 27 scalar cutoffs: accuracy rises 3.46 points and errors fall from 285 to 145. Trained heads learn 27 x (1,024 weights + intercept), reaching 34 errors.

The zero-shot architecture still accepts new questions at runtime, but a new question does not inherit a validated label-specific threshold. Trained heads require labeled data and retraining.

Figure 3. Calibration changes 27 scalar decision thresholds without changing the encoder weights. It cut ModernBERT errors from 285 to 145; trained heads reduced them to 34.

4. Tokenomics: the variables that actually move cost

Define:

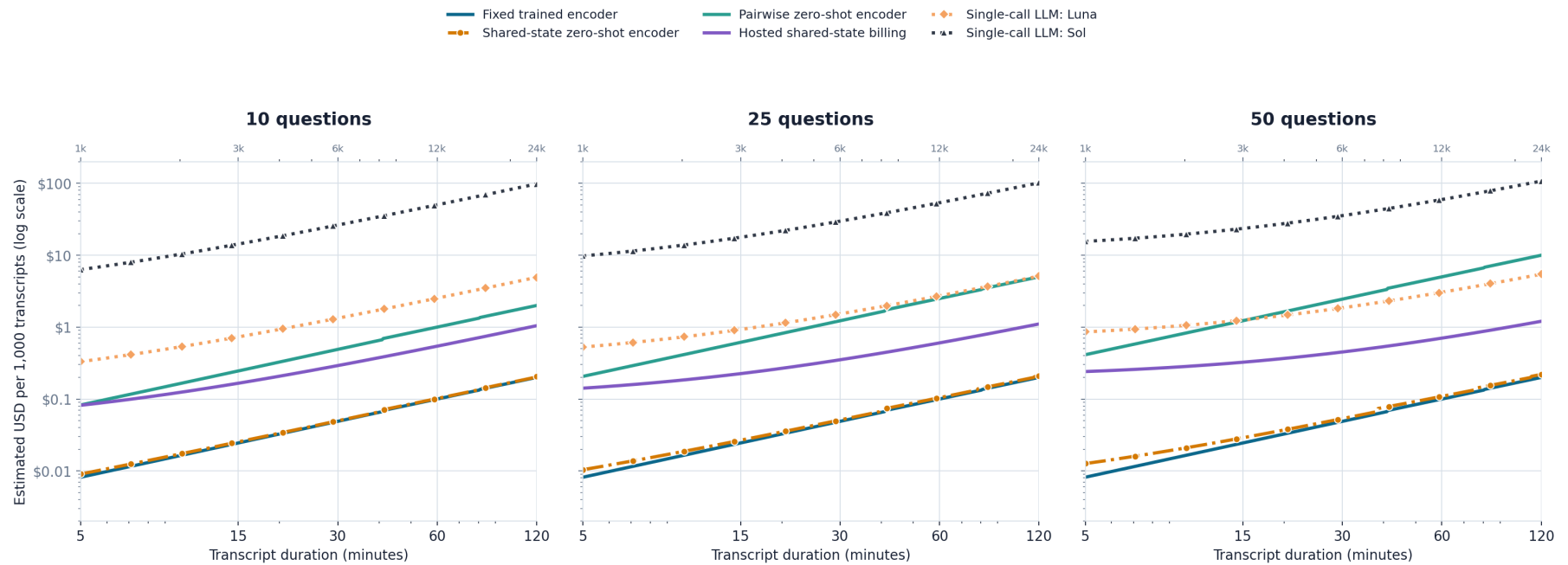
- S : state tokens
- N : number of questions
- U : paid GPU utilization
- R : saturated processed-token throughput
- P : GPU price/hour
- L : compiled labels per pass

Explain cost formulas for pairwise, shared-state, fixed-head, hosted, and LLM systems. Show the 10/25/50-question curves, state-length curves, and utilization crossovers.

FIGURE 4 | CLASSIFIER COST BY STATE DURATION AND TAXONOMY SIZE

Transcript classification cost by duration and taxonomy size

Top axes show state tokens · 200 state tokens/minute scenario · fixed encoder heads versus runtime questions



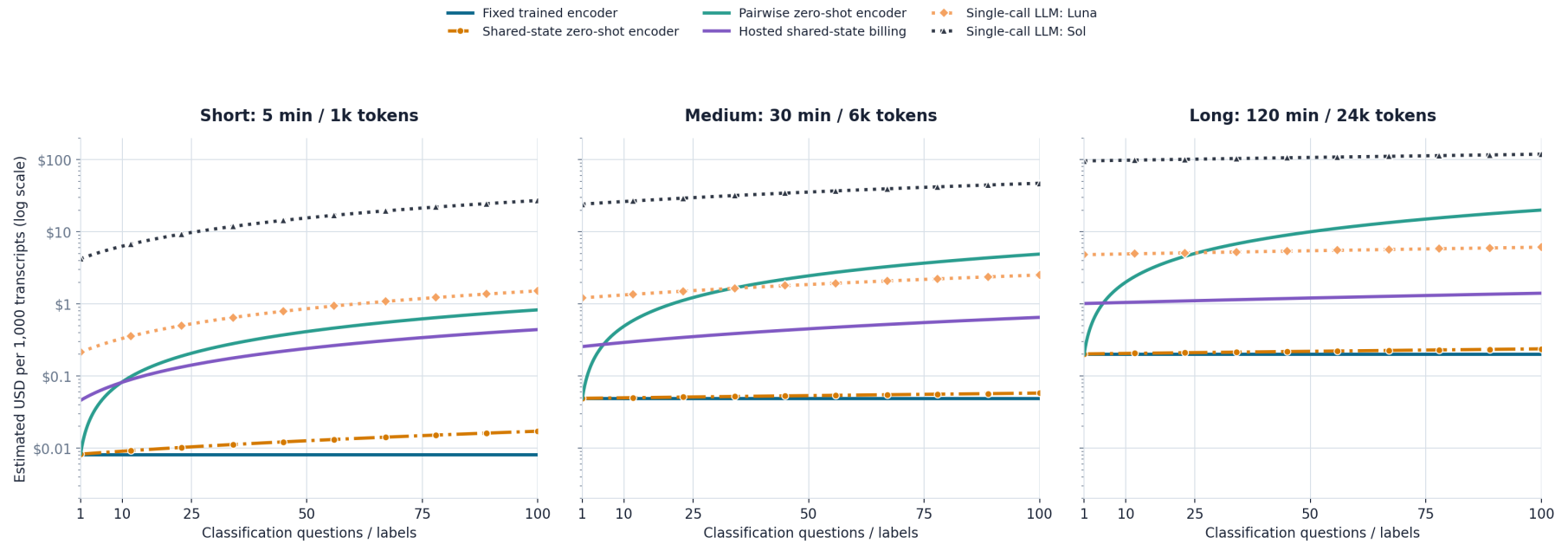
Question overhead is scaled proportionally from the measured 27-question workload. Fixed trained heads require prior training; JEV shows billed tokens, not verified internal compute. Hidden LLM reasoning tokens are excluded.

Figure 4. Cost as the state grows, shown at 10, 25, and 50 questions. The transcript-duration labels are an example; the upper axis is the reusable state-token variable.

FIGURE 5 | CLASSIFIER COST AS THE NUMBER OF RUNTIME QUESTIONS GROWS

Transcript classification cost as the taxonomy grows

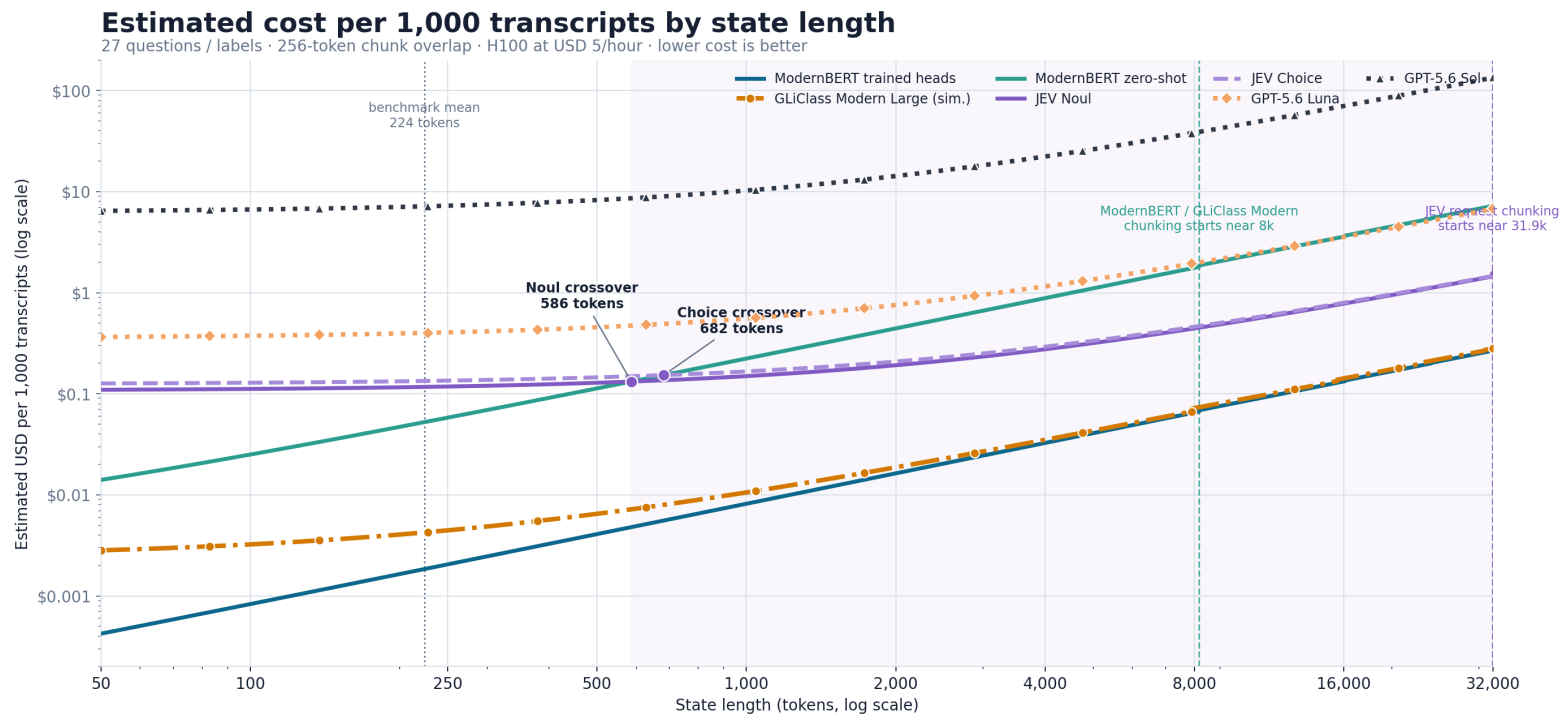
Short, medium, and long transcript scenarios · pairwise NLI repeats the state for every runtime question



Fixed trained-head cost is nearly flat because head compute is treated as negligible. Runtime-label systems scale label, billing, or output overhead without repeating the full state in the external cost model.

Figure 5. Cost as the taxonomy grows for short, medium, and long states. Pairwise NLI becomes increasingly expensive because every new question carries another copy of the state.

FIGURE 6 | DETAILED COST SENSITIVITY BY STATE LENGTH



Each curve is the estimated cost of 1,000 transcripts. Sol/Luna include standardized input plus 212 output tokens; hidden reasoning tokens are excluded. GLiClass Modern uses an H100 throughput proxy, not a measured H100 result. Long-context throughput changes are not modeled.

Figure 6. The detailed 27-question state-length sensitivity, including context-window chunking. This is the bridge between the generic formulas and the measured transcript example.

5. Latency: resource time is not response time

- Model-service time from the H100 throughput proxy.
- Network and API overhead for hosted systems.
- Queueing growth as utilization approaches saturation.
- Interactive p95/p99 versus offline throughput.
- Parallel chunks require additional concurrent capacity; they do not remove compute.

Use the following relationship in the article instead of an accuracy-versus-latency scatter:

```
end-to-end latency = queueing + preprocessing + model service
+ postprocessing + network
```

The model-throughput estimate describes the `model service` component. It is not a p95 or p99 production latency promise.

6. The data scientist's decision path

- Start with labeled examples and the simplest deployable baseline.
- Use hosted APIs or local CPU before buying infrastructure.
- Evaluate quality per label, not only aggregate accuracy.
- Calibrate thresholds on validation data.
- Escalate ambiguous cases to an LLM or human.
- Bring evidence - request rate, state lengths, question counts, burstiness, and SLO - to the platform team.

7. The enterprise platform team's decision path

- Consolidate workloads into a multi-tenant pool.
- Separate stable base load from bursts.
- Pin models and version decision policies.
- Protect interactive capacity with utilization headroom.
- Make data-residency, security, audit, support, and failure-mode costs explicit.
- Use hosted overflow and keep an exit path in both directions.

FIGURE 7 | OPERATING-MODEL SCORECARD

Operating-model scorecard

Named examples from this benchmark · choose the operating model before choosing the model

Operating pattern	Example implementation	Accuracy	Runtime questions	Deployment	Executive takeaway
Fixed trained encoder	ModernBERT-large + trained logistic heads	99.16%	No	Self-hosted	Default economic path when taxonomy is stable
Shared-state zero-shot	GLiClass Modern Large v3	97.90%	Yes	Self-hosted	Flexible winner above ~14.6% utilization at 6k × 25
Hosted shared-state API	TypeSafe.ai JEV Noul	99.90%	Yes	Managed API	Usage-priced; no customer-owned idle GPU
Pairwise zero-shot	ModernBERT-large zero-shot NLI	96.42%	Yes	Self-hosted	Repeats state for every question
Single-call LLM	GPT-5.6 Luna / GPT-5.6 Sol	99.75% / 99.93%	Yes	Managed API	Premium option for broader semantic reasoning

Accuracy is measured on the locked 150-transcript synthetic test. The 14.6% utilization crossover is specific to the 6k-state-token × 25-question, USD 5/H100-hour scenario.

Figure 7. Choose the operating pattern before choosing a particular model. The critical split is fixed versus runtime taxonomy, followed by managed versus self-hosted deployment.

8. When to move off a cloud API

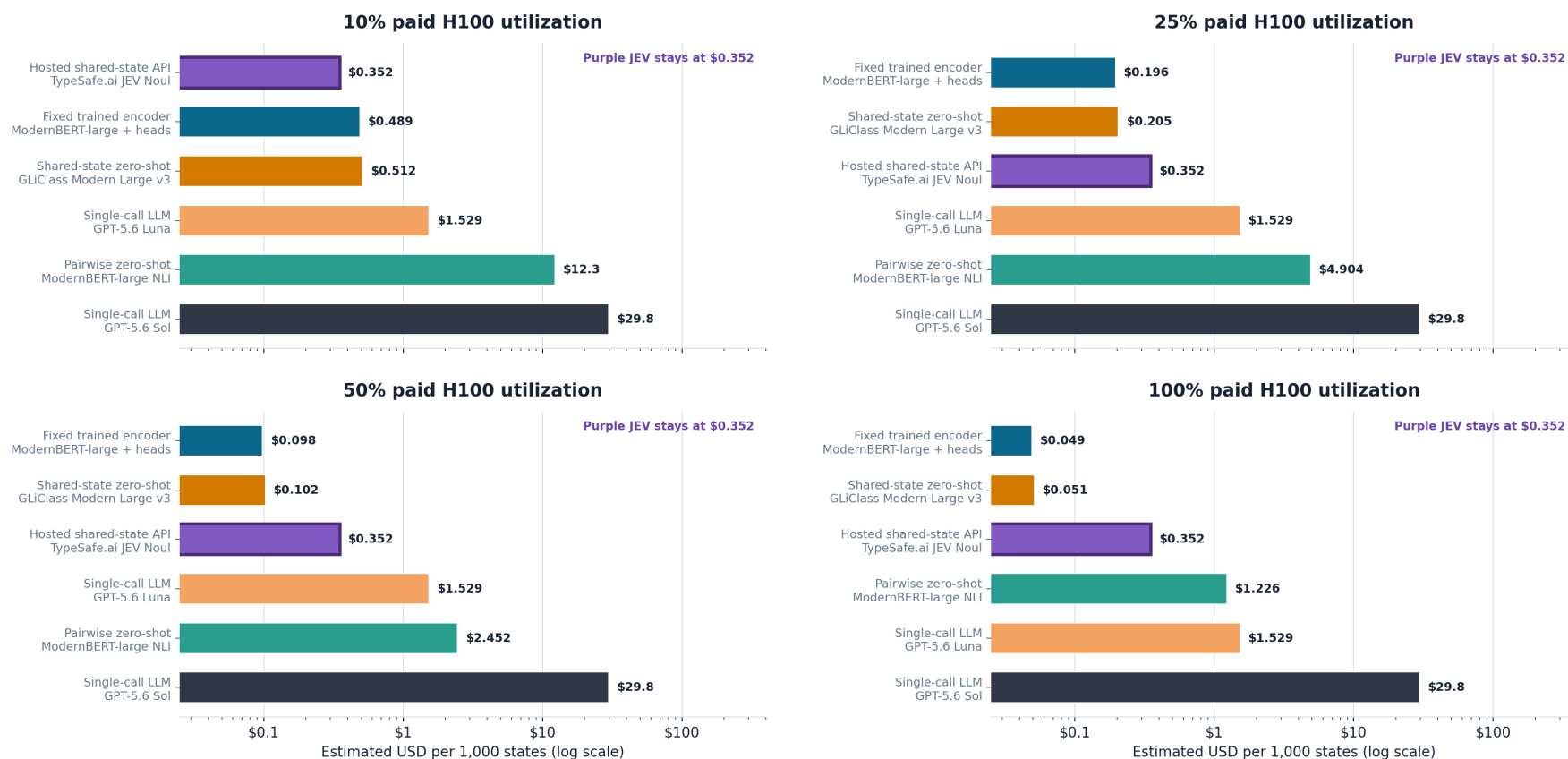
Use a decision gate, not a slogan:

1. Quality parity on a production shadow set.
2. Sustained demand, not peak demand, exceeds the all-in cost crossover.
3. The platform can keep accelerators productively shared.
4. Measured p95/p99 meets the SLO with headroom.
5. Reliability, privacy, and operational ownership are funded.

Illustrate the 6,000-state-token, 25-question reference: GLiClass's raw-inference crossover with JEV is about 14.6% paid utilization, while a prudent operational review would occur at a higher sustained utilization after including platform costs.

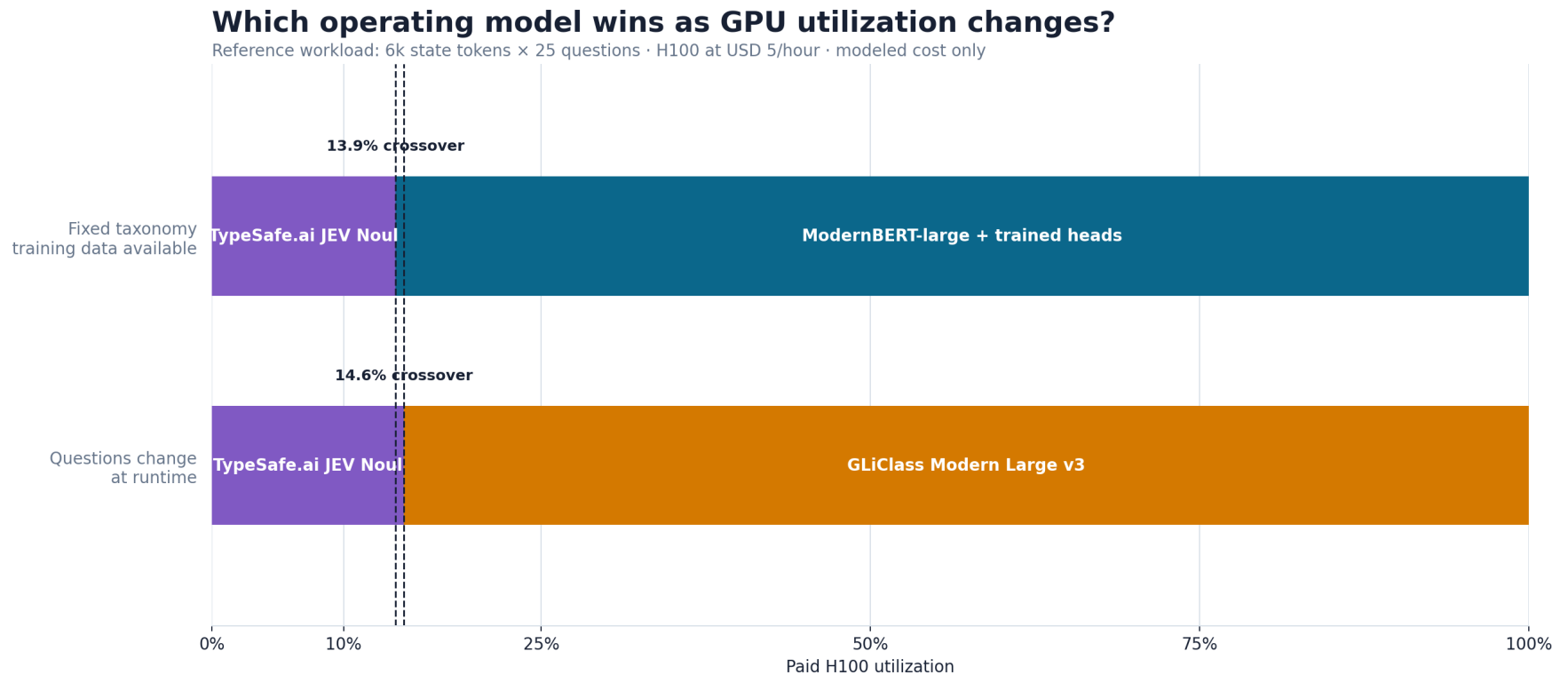
FIGURE 8 | COST PER 1,000 STATES AT FOUR LEVELS OF PAID H100 UTILIZATION**GPU utilization determines self-hosted encoder economics**

Fixed workload: 6k state tokens × 25 questions · NVIDIA H100 at USD 5/hour · lower is better



Self-hosted encoder cost is divided by paid GPU utilization; hosted JEV/Luna/Sol prices remain usage-based. In this scenario, JEV crosses trained ModernBERT near 13.9% utilization and GLiClass Modern near 14.6%. Hidden LLM reasoning tokens are excluded.

Figure 8. Usage-priced JEV remains constant while self-hosted encoder cost falls as paid GPU utilization rises. The bars use a 6,000-token state, 25 questions, and a \$5/H100-hour scenario.

FIGURE 9 | OPERATING-MODEL WINNERS AS UTILIZATION CHANGES

Below the crossover, usage-priced JEV is cheaper than carrying idle H100 capacity. Above it, self-hosted encoders win on modeled marginal cost. Operational requirements and measured quality remain separate gates.

Figure 9. Modeled marginal-cost crossover bands. JEV wins below approximately 13.9% utilization for a fixed taxonomy and 14.6% when questions must change at runtime. Quality and operational readiness remain separate gates.

For readers who need all three variables at once, include the paired decision maps immediately after the crossover discussion:

FIGURE 10 | FIXED-TAXONOMY COST WINNERS

Fixed taxonomy: which approach has the lowest modeled cost?
Trained ModernBERT heads are eligible · state length × taxonomy size × paid GPU utilization



Each cell shows the winner and USD per 1,000 states. ModernBERT-large + trained heads encodes the state once and has negligible incremental head cost, but new or changed questions require labeled data and retraining.

Figure 10. Lowest-cost eligible approach across state length, question count, and utilization when trained heads are allowed.

FIGURE 11 | RUNTIME-QUESTION COST WINNERS

Runtime questions: lowest-cost approach

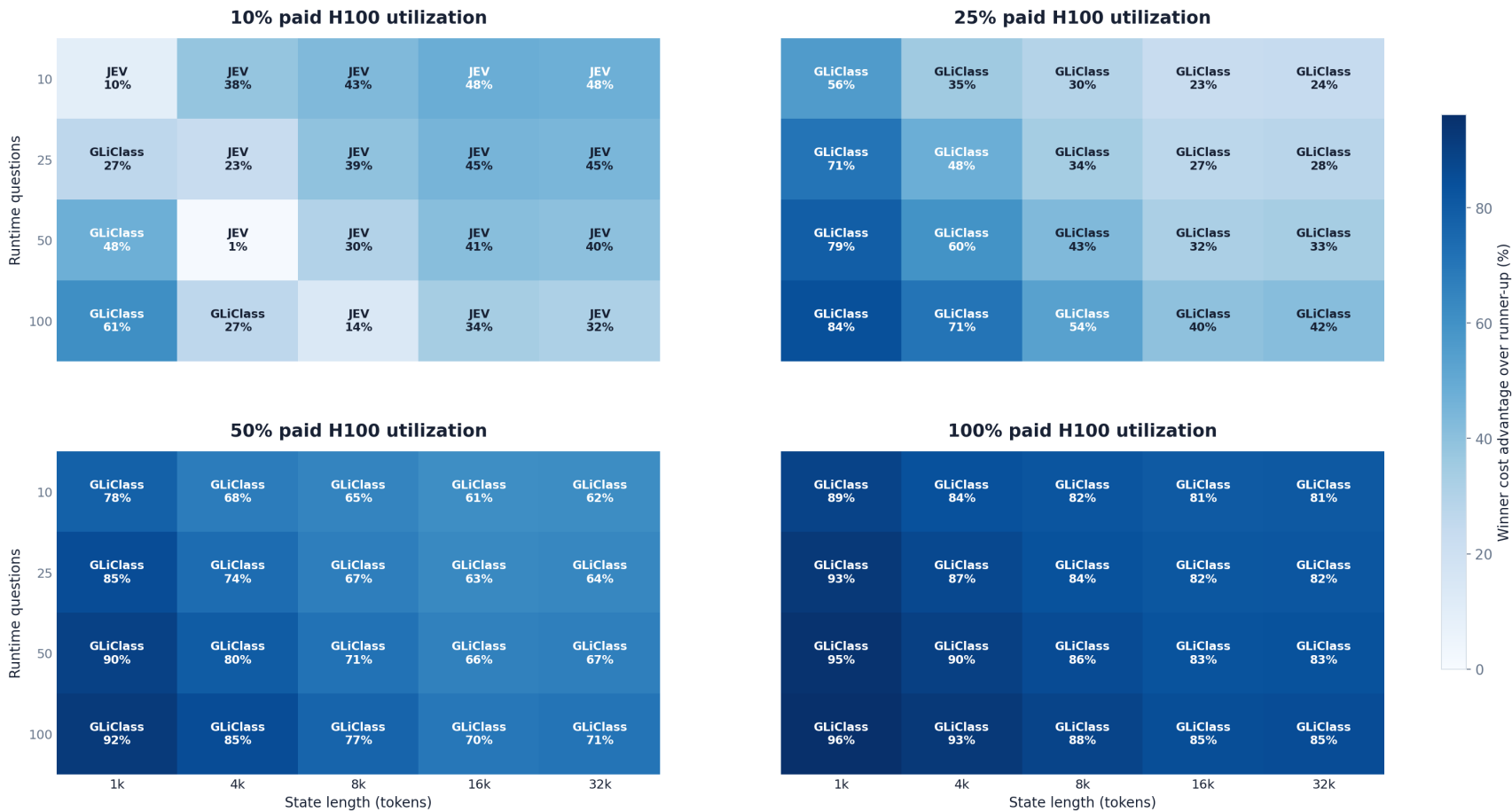
Trained ModernBERT heads are ineligible · state length × question count × paid GPU utilization



FIGURE 12 | SAVINGS ADVANTAGE OF THE RUNTIME-QUESTION WINNER

How decisive is the modeled cost winner?

Runtime questions only; trained heads are ineligible · high percentages are robust, low percentages indicate a close decision



Advantage = (runner-up cost - winner cost) / runner-up cost. Operational simplicity, data residency, latency, support, and quality may outweigh a small modeled cost advantage.

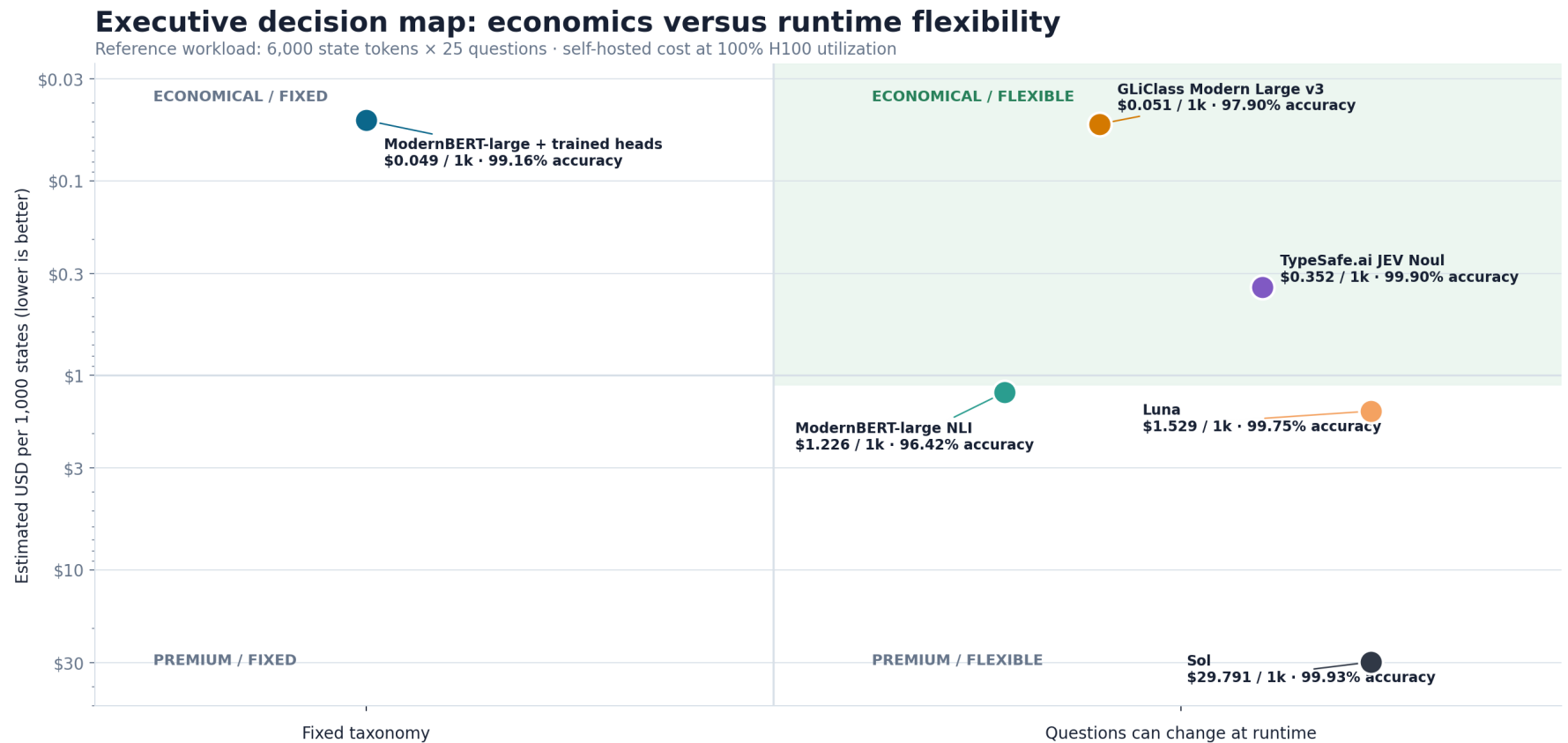
Figure 12. The winning runtime-flexible approach's percentage cost advantage over the runner-up. Low-percentage cells identify close decisions where quality, support, latency, data residency, or operational simplicity may matter more than the modeled savings.

9. What we are testing now

- Whether the strongest open-source shared-state option can provide the same practical state-plus-questions experience.
- How its throughput, latency, calibration, and economics change on production-style hardware.
- Whether one enterprise decision layer can route between trained encoders, runtime-label encoders, JEV, and selective LLM escalation.
- Keep this deliberately brief: the work is underway, and results will follow.

10. A pragmatic reference architecture

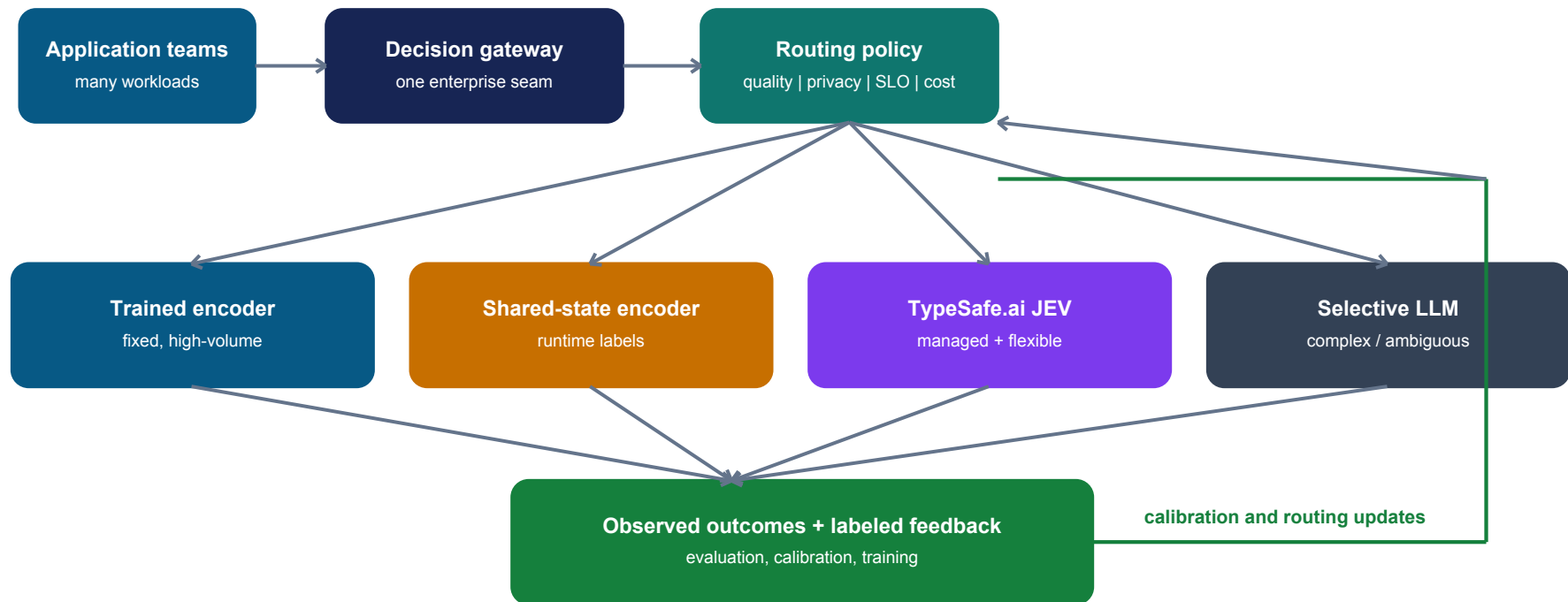
Before the implementation diagram, summarize the available operating positions in one executive view:

FIGURE 13 | EXECUTIVE DECISION MAP OF ECONOMICS AND RUNTIME-QUESTION FLEXIBILITY

The \$1/1,000 boundary is an executive guide, not a universal purchasing threshold. Flexible systems accept new questions without retraining; trained heads do not.

Figure 13. There is no single classifier winner. Trained ModernBERT leads the economical/fixed corner, GLiClass represents economical/runtime-flexible self-hosting, JEV provides a managed flexible option, and LLMs occupy the premium reasoning tier.

REFERENCE ARCHITECTURE



A hybrid decision gateway routes each workload according to quality, privacy, latency, flexibility, and cost, then feeds observed outcomes back into calibration and routing policy.

11. Bottom line

- Fixed taxonomy plus sustained scale: trained heads are difficult to beat.
- Runtime questions plus efficient self-hosting: shared-state zero-shot encoders are compelling.
- Variable or early demand: usage-priced JEV avoids utilization risk.
- Complex reasoning or ambiguity: use an LLM selectively, not automatically.
- The winning enterprise design is usually hybrid and instrumented.

Visual editorial notes

- Lead with the architecture picture, not a benchmark leaderboard; the article is a decision guide.
- Use F1 as the main quality chart and the calibration chart as the practical intervention.
- Keep both generic cost charts because they isolate the two variables readers control: state length and number of questions.
- Keep the detailed state-token chart after the generic charts, where technical readers can inspect chunking effects.
- Put the utilization bars and crossover bands in the cloud-versus-self-hosting section.
- Put the two heatmaps together; they are a paired comparison and use the same axes.
- Follow the runtime winner map with the savings-advantage heatmap so readers can distinguish decisive wins from near ties.
- Use the 2x2 matrix immediately before the reference architecture as the executive synthesis.
- Do not use `estimated-cost-1000-transcripts.png` in the article because it compares API charges with local models whose hardware cost is shown as zero.
- Do not use `accuracy-vs-equivalent-latency.png`; the measurements mix hosted round trips, local CPU runs, and H100 resource-time estimates.
- Keep `accuracy-comparison.png` and `all-metrics-table.png` as optional supporting figures for the technical report rather than the main article.